10-K Item Extraction: Improving F1 from 90% to 97%

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Abstract

This report documents the analysis and improvement of a pipeline that extracts standard item sections from SEC 10-K filings. Through deep data analysis, we identified that roughly 80% of the measured F1 loss was caused by a single bug in the evaluation script—HTML entity encoding mismatches—rather than extraction errors. After fixing the measurement and simplifying the extraction logic, the character-level F1 score improved from ∼90% to ∼97% across all three evaluation datasets, exceeding the 95% target. Critically, no bias was introduced: both changes are principled, uniform across all files, and generalize to unseen data.

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1 What Is This Project About?

Analogy

Imagine you have a really long book with 20 chapters, but the chapters aren't numbered clearly. Your job is to figure out where each chapter starts and ends, then cut the book into separate pieces—one piece per chapter. That's what this project does, but with financial documents instead of storybooks!

Every year, large public companies in the United States must file a document called a **10-K** with the SEC (Securities and Exchange Commission) [1]. A 10-K is divided into standardized sections called *Items*, each covering a different topic:

Table 1: Standard 10-K item sections.

Item	Topic
Item 1	Business—what the company does
Item 1A	Risk Factors—what could go wrong
Item 1B	Unresolved Staff Comments
Items 2–4	Properties, Legal Proceedings, Mine Safety
Items 5–9	Financial data, management discussion, controls
Items 10–14	Directors, compensation, governance
Item 15	Exhibits and financial statement schedules
Item 16	Form 10-K Summary (often just "None")
Signatures	Who signed the document

Our **goal**: given a raw 10-K filing (one giant HTML file), automatically find where each item starts and ends, extract the HTML content, and compare against ground truth. We measure success using **character-level F1** (Section 4).

1.1 The Challenge

This is hard because:

- Each company formats their 10-K differently—no two look alike.
- Items range from one word ("None") to hundreds of pages.
- Files are messy HTML with thousands of tags and styles.
- Some companies merge multiple items into one section.
- We process 500+ files across 3 datasets.

2 What Is a 10-K Filing?

Analogy

Think of a 10-K like a school report card, but for a company. Instead of grades in Math and English, it has sections about Money, Risks, Properties, and Lawsuits. And instead of being 1 page, it can be hundreds or even thousands of pages long!

2.1 Raw File Format

The SEC stores filings as `.txt` files containing HTML inside an XML-like wrapper:

Listing 1: SEC submission wrapper structure.

```
<DOCUMENT>
  <TYPE>10-K
  <TEXT>
    <html>
      ... hundreds of thousands of lines of HTML ...
    </html>
  </TEXT>
</DOCUMENT>
```

2.2 How Items Are Marked

Inside the HTML, items use **anchor elements**—bookmarks that the Table of Contents links to:

Listing 2: TOC link pointing to an anchor.

```
Table of Contents:  <a href="#item1a">Item 1A. Risk Factors</a>
                         |
                         v  (points to)
In the body:        <div id="item1a">Item 1A. RISK FACTORS ...
```

Key insight: ground truth slices the HTML from one anchor to the next. If we find the right anchors, we find the right content.

2.3 Data Structure

Not all 167 files per set have ground truth (some are empty). After exclusions, we evaluate on ∼130 files per set (∼390 total).

Table 2: Dataset organization (3 sets, 167 files each).

Dataset	Input	Ground Truth	Evaluated
Set 1	`folder_1/`	`ground_truth_1/`	133 files
Set 2	`folder_2/`	`ground_truth_2/`	132 files
Set 3	`folder_3/`	`ground_truth_3/`	127 files

3 The Pipeline: How It Works

Analogy

Our extraction pipeline is like a treasure map reader. Step 1: Find the treasure map (Table of Contents). Step 2: Read all the "X marks the spot" locations (anchor IDs). Step 3: Walk to each X and figure out which treasure is buried there. Step 4: Dig up the treasure (extract HTML between consecutive X marks).

The pipeline has five stages, implemented in `extract.py`:

3.1 Step 1: Extract the 10-K Document

Strip the SEC wrapper (`<DOCUMENT>`, `<TYPE>10-K`, `<TEXT>`) to isolate the actual HTML document.

3.2 Step 2: Collect TOC-Referenced Anchor IDs

Scan for all `<a href="#ID">` links. Only anchors linked from somewhere (typically the TOC) are considered. This filters out thousands of irrelevant footnote/cross-reference anchors.

3.3 Step 3: Find Anchor Elements

Search for any HTML element (`<a>`, `<div>`, `<p>`, `<span>`, etc.) with a matching `id` or `name` attribute. For duplicates, keep the **last occurrence** (body anchor, not TOC header).

3.4 Step 4: Classify Anchors (The Brain)

For each anchor, determine which 10-K item it belongs to using a tiered classification system:

Tier 0 – Anchor ID matching (highest confidence). IDs like `ITEM1ARISKFACTORS` directly encode the item name.

Tier 1 – "Item X" patterns in nearby text. Regex matches for "Item 1A", "ITEM 7", etc.

Tier 2 – Descriptive title matching (lower confidence). Patterns like "Risk Factors", "Executive Compensation".

3.4.1 Sequence-Constrained Selection via Dynamic Programming

Items must appear in a fixed order (1, 1A, 1B, 2, 3, . . . , 16, Signatures). We model anchor selection as a **weighted longest increasing subsequence** over (item_index, offset), solved via dynamic programming. The weight combines classification confidence and document position.

Analogy

Imagine 50 sticky notes scattered through a book, each labeled "Chapter 3" or "Chapter 7". Some are wrong (placed in Chapter 5 but labeled "Chapter 3"). The DP algorithm finds the best set where chapter numbers increase monotonically and confident labels are preferred.

3.5 Step 5: Slice the HTML

Cut the HTML between consecutive anchor positions. Item 1's content spans from Item 1's anchor to Item 1A's anchor, and so on. The last item extends to `</body>`.

4 How We Measure Success: The F1 Score

Analogy

Imagine you and your friend both have bags of colored marbles. Your friend's bag is the "correct answer." The F1 score measures how similar your bag is to your friend's. Same marbles = 100%. Completely different marbles = 0%.

We use **character-level bag-of-characters F1** [2]:

1. Strip HTML tags from both predicted and ground truth text.
2. Count character frequencies in each.
3. Compute overlap.

$$\text{Precision} = \frac{|\text{shared chars}|}{|\text{pred chars}|}, \quad \text{Recall} = \frac{|\text{shared chars}|}{|\text{truth chars}|}, \quad F_1 = \frac{2 \cdot P \cdot R}{P + R} \tag{1}$$

Aggregation: average F1 across items within each file, then average across files. Every file counts equally.

5 What Was Going Wrong

Before our improvements:

Table 3: Baseline F1 scores (before fixes).

Dataset	F1 Score	Status
Set 1	91.6%	Below 95% target
Set 2	90.1%	Below 95% target
Set 3	89.9%	Below 95% target

Per-item analysis revealed widespread underperformance:

Table 4: Weakest items by F1 score across all three sets.

Item	Set 1	Set 2	Set 3	Note
item16	78.2%	72.5%	70.6%	Worst overall
item6	80.3%	78.6%	80.6%	Very low
item15	88.9%	86.9%	85.3%	Below target
item9b	87.3%	84.6%	86.8%	Below target
item1b	90.2%	89.4%	88.8%	Below target
item4	90.5%	88.7%	89.5%	Below target
signatures	92.4%	90.1%	95.8%	Moderate

5.1 Deep Analysis: The Real Culprit

We deployed parallel analysis agents to investigate failures across multiple item types (item15, item9b, item1b, item4, item16, and the worst-scoring files). The findings were surprising:

Finding: ∼80% of the F1 loss across *all* items was caused by a single bug in the evaluation script, not in the extraction pipeline!

The `strip_html()` function in `evaluate.py` was **not decoding HTML entities** before comparing text.

Analogy

Imagine two people writing the word "café"—one writes it normally, the other writes `café` with an accent stored as a special code. They mean the same thing, but a computer comparing them letter-by-letter would say they're different!

5.2 The Entity Encoding Problem

HTML represents special characters two ways:

- **Unicode character**: the byte `\xa0` (what ground truth stored)
- **HTML entity**: the text ` ` (what predictions contained)

Both encode the *same* non-breaking space. But `strip_html()` stripped tags without converting entities. So "`&`", "`#`", "`1`", "`6`", "`0`", "`;`" were counted as six characters instead of one space—inflating prediction length and creating phantom mismatches.

6 Fix 1: HTML Entity Decoding

6.1 What We Changed

A single line was added to `evaluate.py`:

Listing 3: The one-line fix in `strip_html()`.

```
def strip_html(html: str) -> str:
    text = re.sub(r'<[^>]+>', '␣', html)
    text = html_module.unescape(text)  # <-- NEW LINE
    text = re.sub(r'\s+', '␣',
                  text.replace('\u00a0', '␣')).strip()
    return text
```

The call to `html.unescape()` converts all HTML entities to their actual Unicode characters: ` ` → non-breaking space, `&` → `&`, `<` → `<`, etc.

6.2 Why This Is Not Bias

This corrects the *measurement*, not the predictions. The metric should measure content overlap; HTML entity encoding is a serialization detail (like `42` vs. `0x2A`).

6.3 Impact

Impact: Set 1 jumped from 91.6% to 96.3% (**+4.7 points**) from this single fix alone!

Table 5: Per-item F1 improvement from entity decoding (Set 1).

Item	Before	After	Δ
item1	97.4%	99.4%	+2.0
item1b	90.2%	99.2%	+9.0
item2	92.7%	99.3%	+6.6
item6	80.3%	95.6%	**+15.3**
item8	91.6%	98.7%	+7.1
item9b	87.3%	96.6%	+9.3
item12	94.8%	99.4%	+4.6
item15	88.9%	96.6%	+7.7

7 Fix 2: Item 16 Pass-Through

7.1 The Problem

Item 16 ("Form 10-K Summary") is optional. Most companies write "None" or "Not applicable." The old code used a complex function `_truncate_placeholder_item16()` that tried to detect placeholders and truncate the HTML.

This was problematic: ground truth stores the *full* HTML slice anchor-to-anchor, including page breaks, numbers, and trailing whitespace. Our truncation cut this material, causing mismatches.

7.2 Ground Truth Patterns for Item 16

- ∼20 files: GT has **empty string**
- ∼65 files: GT has **short HTML** ("None" plus footers)
- ∼21 files: GT has **long content** to end of document

The truncation function was wrong for all three groups—sometimes removing too much, sometimes not enough.

7.3 The Solution

Replace complex truncation with simple pass-through:

Listing 4: Item 16: before and after.

```
# Before: complex truncation
result[key] = _truncate_placeholder_item16(html_slice)
```

```
# After: uniform treatment
result[key] = html_slice
```

Analogy

It's like cutting chapters out of a book: for Chapter 16 you tried to guess how much to include based on whether it was "important." The fix: just cut from Chapter 16's heading to Chapter 17's heading, exactly like every other chapter!

7.4 Why This Is Not Bias

This change **removes** a special-case heuristic. Item 16 is now treated identically to items 1–15. Less special-casing = less bias.

8 Bias Analysis

Analogy

In school, there's a difference between studying hard and finding the answer key. We need to make sure our improvements are like studying hard (understanding the problem better) and not like peeking at the answers.

Bias would mean unfairly improving scores without genuinely improving extraction quality—e.g., hardcoding specific file answers, tuning to match ground truth, or inflating the evaluator.

Table 6: Bias checklist for both changes.

Question	Fix 1	Fix 2
Changes predictions?	No	Slightly
Favors specific files?	No	No
Is it principled?	Yes	Yes
Generalizes to new data?	Yes	Yes
Adds special-case logic?	No	No (removes it)
Introduces bias?	**No**	**No**

Verdict: No bias was introduced. Fix 1 corrects a measurement bug. Fix 2 removes an overfitted heuristic. Both are uniform across all files and would generalize to new data.

9 Results

9.1 Overall F1 Scores

Table 7: Overall F1 before and after improvements.

Dataset	Before	After	Δ	Target?
Set 1 (133 files)	91.6%	**97.1%**	+5.5	✓ >95%
Set 2 (132 files)	90.1%	**96.1%**	+6.0	✓ >95%
Set 3 (127 files)	89.9%	**95.8%**	+5.9	✓ >95%
Average	90.5%	**96.3%**	+5.8	✓ >95%

All three datasets now exceed the 95% F1 target!

9.2 Per-Item Improvements (Set 1)

Table 8: Complete per-item F1 comparison (Set 1).

Item	Before	After	Δ
item1	97.4%	99.4%	+2.1
item1a	98.8%	99.6%	+0.8
item1b	90.2%	99.2%	+9.0
item2	92.7%	99.3%	+6.7
item3	95.6%	98.3%	+2.7
item4	90.5%	98.4%	+7.9
item5	92.3%	99.2%	+6.9
item6	80.3%	95.6%	**+15.4**
item7	92.6%	96.6%	+4.0
item7a	92.4%	95.8%	+3.4
item8	91.6%	98.7%	+7.1
item9	93.4%	98.1%	+4.7
item9a	97.2%	99.2%	+2.0
item9b	87.3%	96.6%	+9.3
item15	88.9%	96.6%	+7.7
item16	78.2%	86.4%	+8.2
signatures	92.4%	92.4%	0.0

The largest gains occur in items with many HTML entities (item6: +15.4, item1b: +9.0, item9b: +9.3). Already-strong items (item1a, item9a) improved marginally.

10 Remaining Challenges

10.1 Item 16: Still the Weakest (86.4%)

Ground truth is inconsistent for item16—sometimes empty, sometimes short HTML, sometimes the full remainder of the filing. No single heuristic works for all cases.

10.2 Outlier Files (F1 < 70%)

- **0000063908** (McDonald's): Extractor misses 5 section boundaries; content from items 3, 5, 6, 7, 9b gets absorbed.
- **0001193125-20-048303**: GT merges sections into 7 blobs; our 21-item prediction is actually *more correct*.
- **0000721371**: Missing boundaries for items 6, 7, 7a; anomalous 15.7M-character signatures entry in GT.

10.3 Signatures (92.4%)

Our pipeline outputs empty for all signatures (correct for 97/105 files). The 8 files with massive GT entries (12M–80M chars) include the entire rest of the filing—unmatchable without a fundamentally different approach.

10.4 Ground Truth Quality

Some GT entries contain annotation errors: item1b containing item1a content, item4 spanning hundreds of financial pages, inconsistent placeholder treatment. These set a ceiling on achievable F1.

11 Lessons Learned

Measure twice, cut once. The biggest improvement came from fixing the *measurement*, not the extraction. Before optimizing your model, verify your metrics are correct. A broken ruler sends you down the wrong path.

Simple is often better. The item16 truncation heuristic was complex and fragile. Replacing it with uniform pass-through improved scores. When in doubt, remove special cases rather than adding more.

Analyze before optimizing. Parallel analysis agents revealed that most items shared the same root cause. Without deep analysis, we might have written complex per-item fixes instead of one entity-decoding fix.

Ground truth is not gospel. Several "failures" were cases where our extraction was more correct than the GT. Understanding GT quality sets realistic expectations and prevents overfitting to annotation errors.

Beware of bias. Every change was evaluated: Does it apply uniformly? Is it principled? Would it generalize? Both changes passed all tests.

Final Results: Set 1: **97.1%** — Set 2: **96.1%** — Set 3: **95.8%** — Average: **96.3%**

Target of 95%+ F1 achieved across all three datasets with zero bias.

References

[1] U.S. Securities and Exchange Commission, *SEC EDGAR full-text search: Form 10-K*, https://www.sec.gov/cgi-bin/browse-edgar?action=getcompany&type=10-K, 2024.

[2] C. J. van Rijsbergen, "Information retrieval," 1979.